

A factored form makes zeros and multiplicities visible:

$$f(x) = a(x - r_1)^{m_1}(x - r_2)^{m_2} \cdots (x - r_k)^{m_k}.$$

1. Determine the degree and leading term; describe both ends.
2. Find the x -intercepts and the y -intercept $(0, f(0))$.
3. Use multiplicity to decide whether the graph crosses or touches each zero.
4. Use a test value between consecutive zeros to determine the sign there.
5. Draw a smooth curve. Check the turning-point bound and both ends.

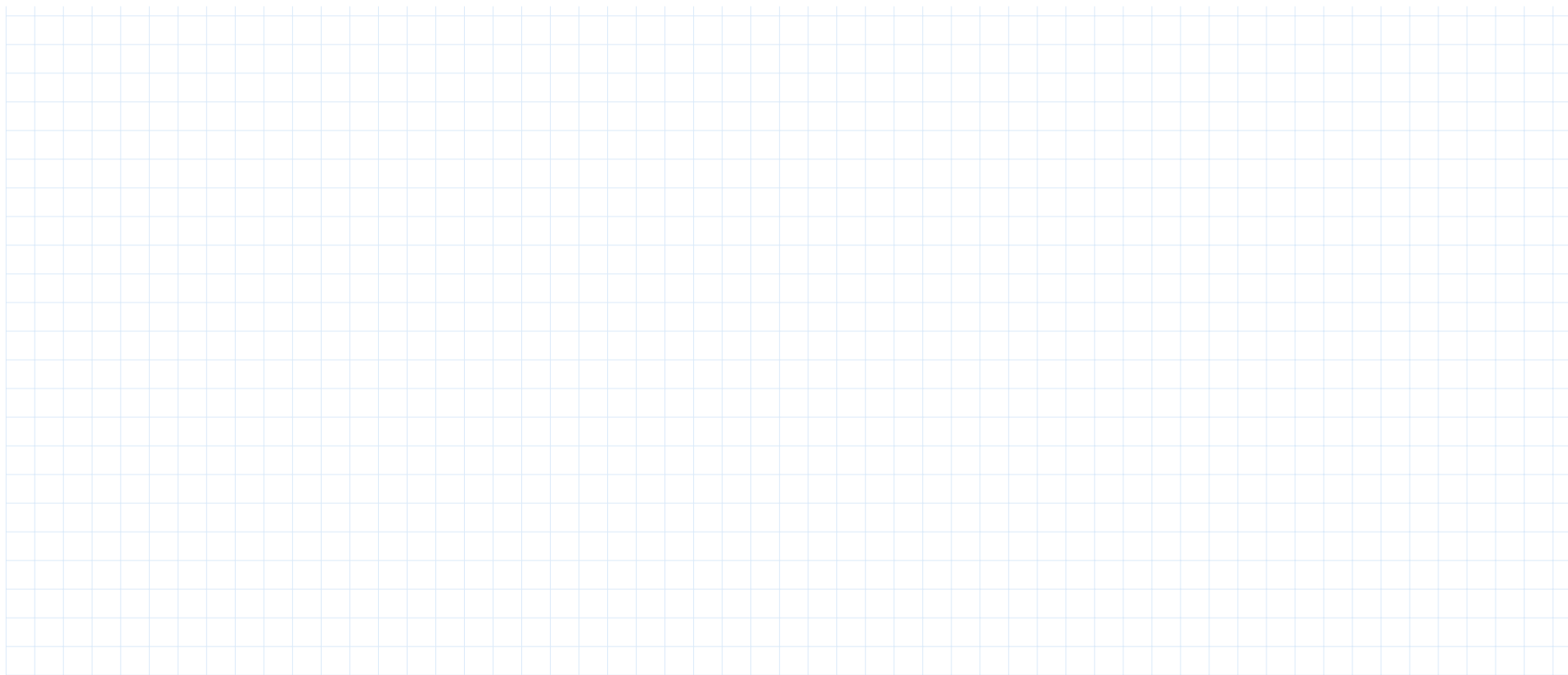
A qualitative sketch should show the correct intercepts, signs, and end behavior.

Exact turning-point coordinates are not required in this section.

Analyze and Sketch a Cubic

Example. Analyze $f(x) = (x + 1)(x - 2)^2$ and sketch its graph.

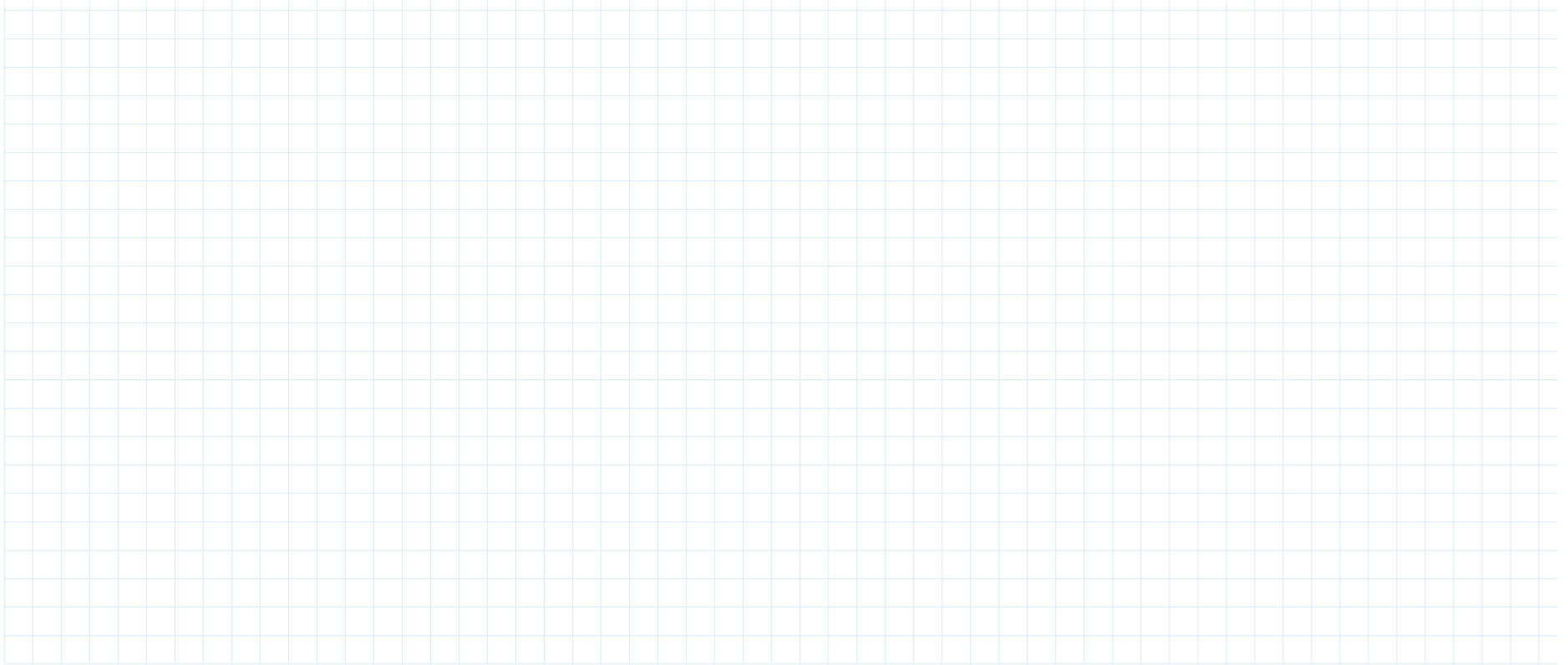
Find the degree and leading term, both end limits, all intercepts, each zero's multiplicity, and the maximum possible number of turning points. Show crossing or touching at each zero.



Two Touching Zeros

Example. Analyze $g(x) = -(x + 2)^2(x - 1)^2$ and sketch its graph.

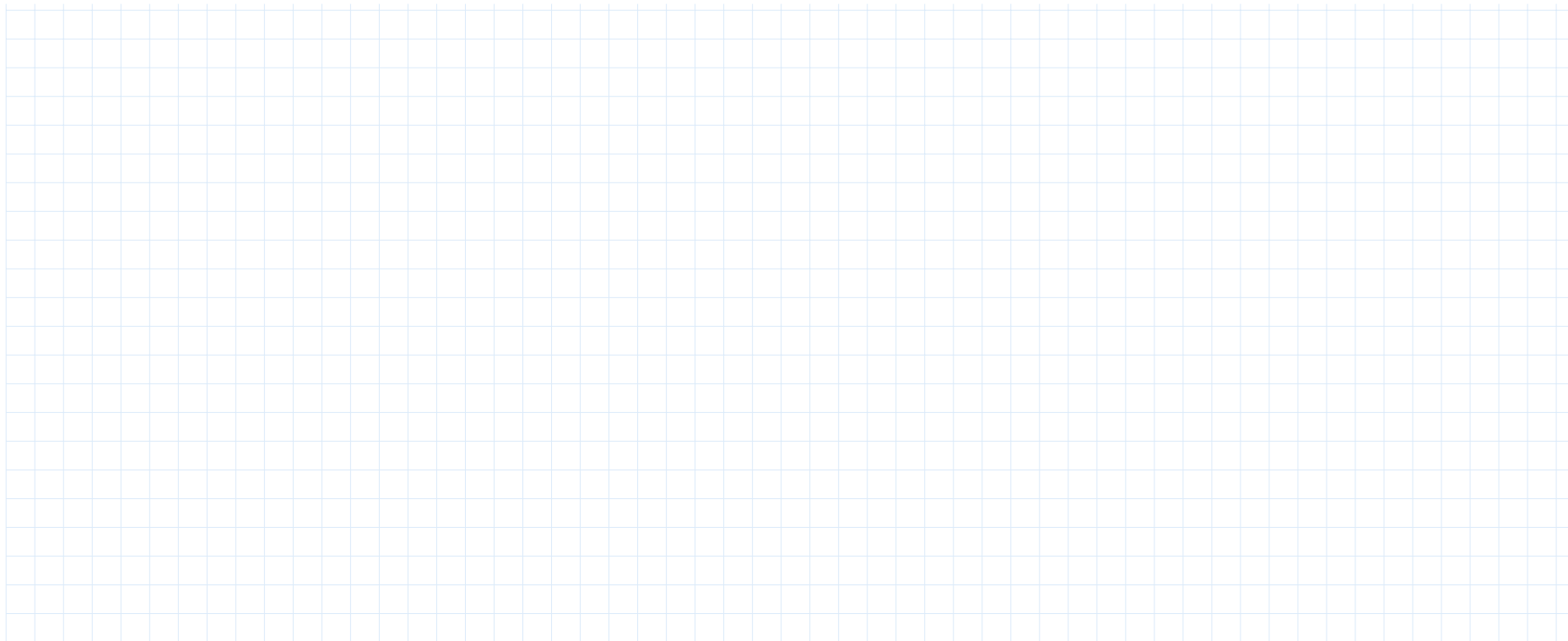
Give the degree, leading term, both end limits, intercepts, and multiplicities. Explain why the graph cannot be above the x -axis. State the turning-point bound.



Build a Polynomial from Zeros

Example. Find the polynomial of least degree with a simple zero at -2 , a double zero at 1 , and passing through $(0, 4)$.

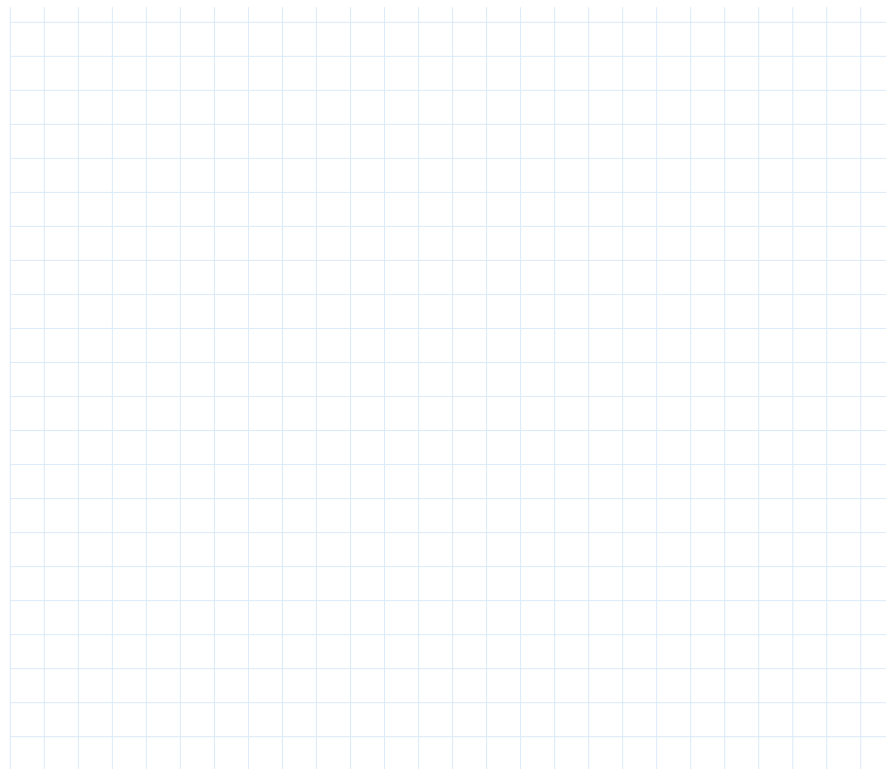
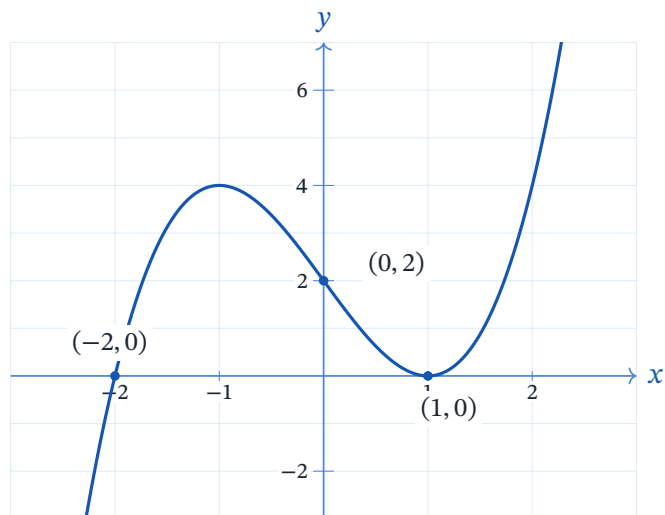
Write its factored form. Show how the given point determines the coefficient a . Then state its degree and leading coefficient.



Suggested lesson break: continue with the graph-to-formula example next class.

Build a Polynomial from a Graph

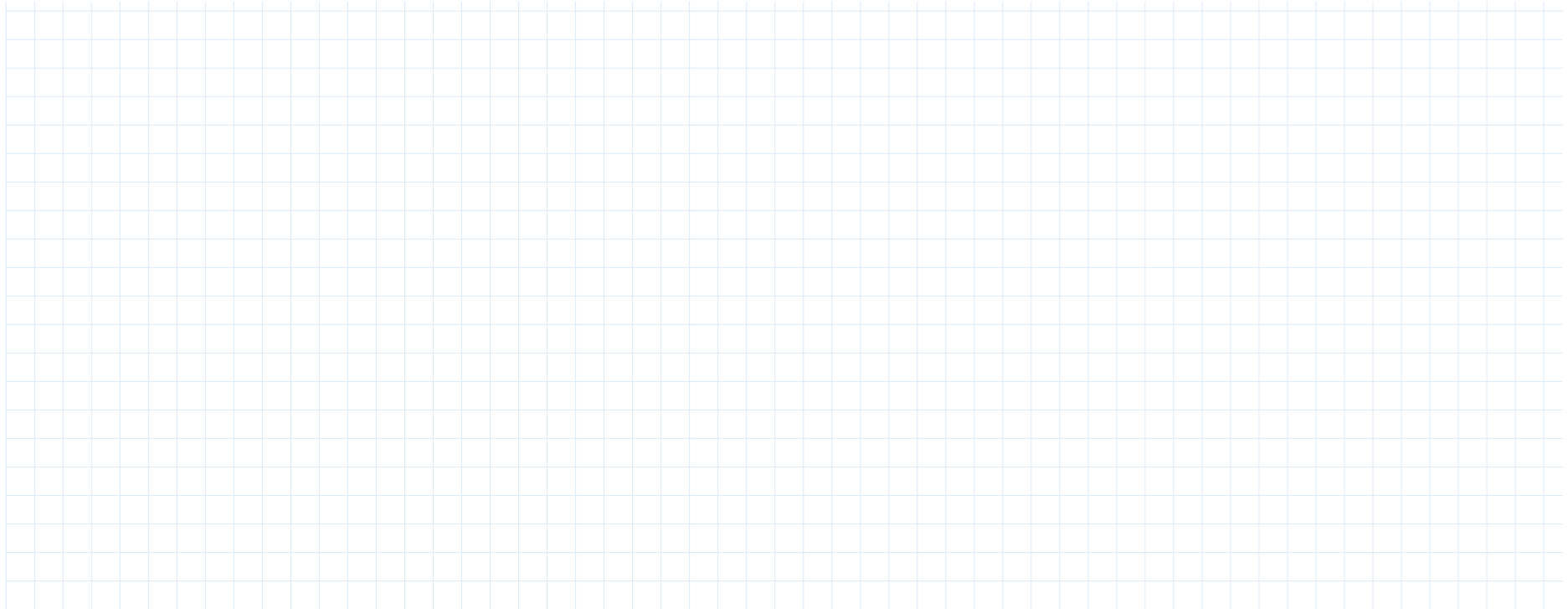
Example. Find a polynomial of least degree that matches the graph. The labeled intercepts are exact. Show how you determine the factors and coefficient.



What Do the Zeros Determine?

Example. A polynomial has simple zeros at -3 , 0 , and 2 .

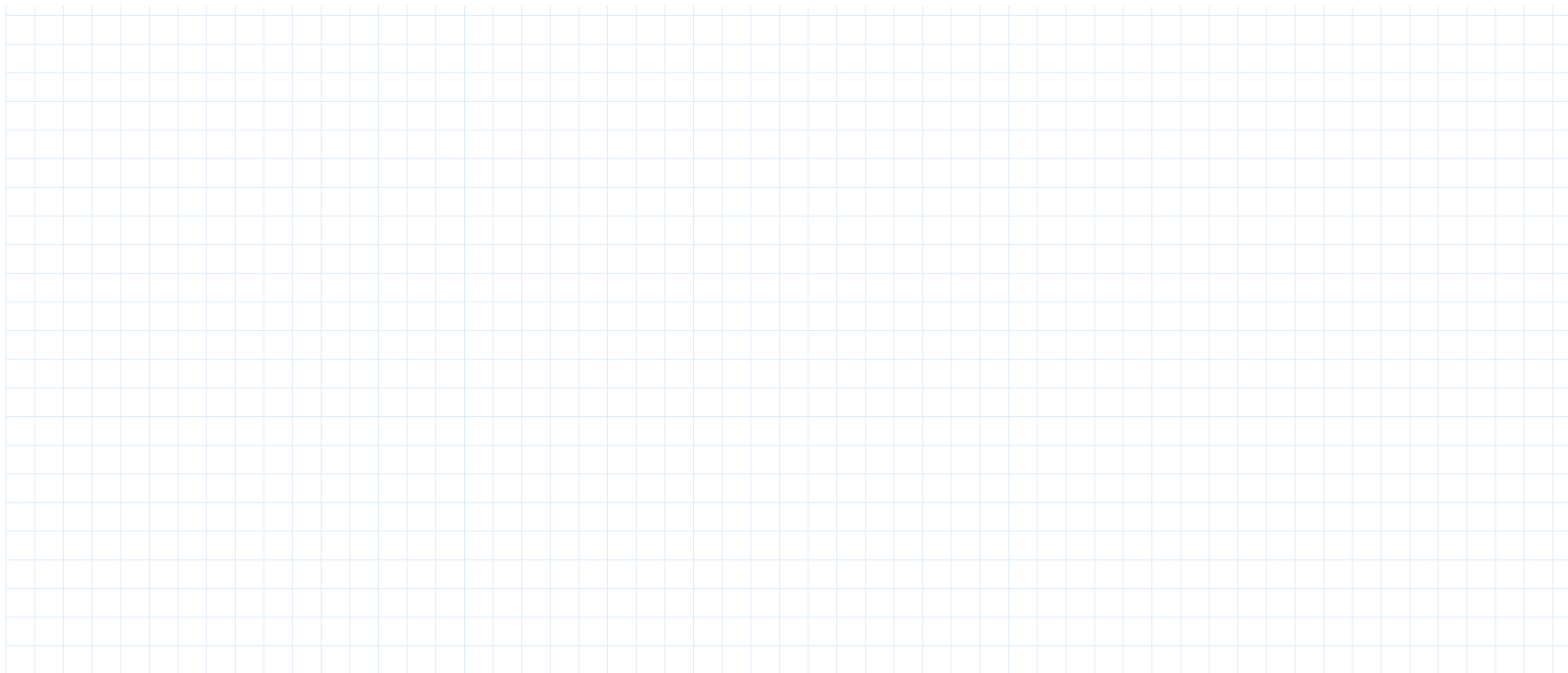
- (a) Write the family of all degree-3 polynomials with these zeros. Is there only one?
- (b) Find the member with leading coefficient -2 ; write it in standard form.
- (c) State its y -intercept and describe both ends with limits.



Factor First, Then Sketch

Example. Analyze $h(x) = x^3 - 4x$ and sketch its graph.

Factor completely. Find all intercepts, multiplicities, both end limits, and the maximum possible number of turning points. Use signs between zeros to guide your sketch.



Build, Check, and Sketch

Example. Find a polynomial p of least degree with a double zero at -1 , a simple zero at 3 , and passing through $(0, -6)$.

Give its factored form, degree, y -intercept, both end limits, and turning-point bound. Sketch a graph consistent with these facts.

